

First Lab

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Part 1: D Flip-Flop FSM Implementation (Moore)

Boolean Equations

$$D_1 = w\overline{Q_1}Q_0 + wQ_1\overline{Q_0}$$

$$D_0 = \overline{w} + Q_1\overline{Q_0}$$

$$y = Q_1Q_0$$

Excitation & State Table

inp	P.S.		Excitation inputs		N.S.		out
w	Q_1	Q_0	D_1	D_0	Q_1^+	Q_0^+	y
0	0	0	0	1	0	1	0
0	0	1	0	1	0	1	0
0	1	0	0	1	0	1	0
0	1	1	0	1	0	1	1
1	0	0	0	0	0	0	0
1	0	1	1	0	1	0	0
1	1	0	1	1	1	1	0
1	1	1	0	0	0	0	1

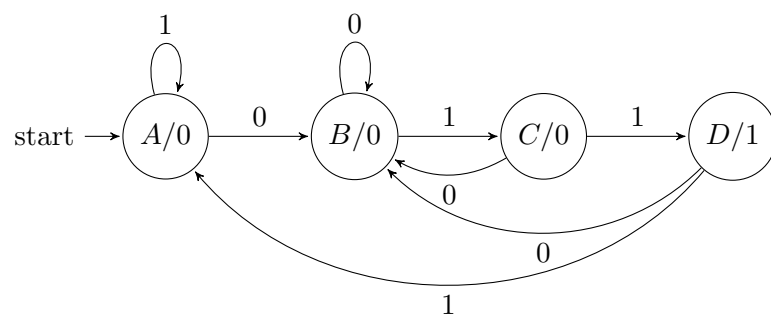
State Encoding Mapping

Q_1	Q_0	State
0	0	A
0	1	B
1	0	C
1	1	D

State Transition Table

w	P.S.	N.S.	out (y)
0	A	B	0
0	B	B	0
0	C	B	0
0	D	B	1
1	A	A	0
1	B	C	0
1	C	D	0
1	D	A	1

State Diagram



Part 2: JK Flip-Flop FSM Implementation (Mealy)

Boolean Equations

$$J_1 = wQ_0$$

$$K_1 = Q_0$$

$$J_0 = w \oplus Q_1$$

$$K_0 = 1$$

$$y = wQ_1Q_0$$

JK Excitation Table

inp	P.S.		Excitation inputs				N.S.		out
w	Q_1	Q_0	J_1	K_1	J_0	K_0	Q_1^+	Q_0^+	y
0	0	0	0	0	0	1	0	0	0
0	0	1	0	1	0	1	0	0	0
0	1	0	0	0	1	1	1	1	0
0	1	1	0	1	1	1	0	0	0
1	0	0	0	0	1	1	0	1	0
1	0	1	1	1	1	1	1	0	0
1	1	0	0	0	0	1	1	0	0
1	1	1	1	1	0	1	0	0	1

State Transition Table

w	P.S.	N.S.	y
0	A	A	0
0	B	A	0
0	C	D	0
0	D	A	0
1	A	B	0
1	B	C	0
1	C	C	0
1	D	A	1

State Diagram

